	Procedural Programming	OOPS
1.	Programs is divided into parts called functions	Program is divided into objects
2.	Doesn't provide proper way to hide data gives importance to function and data moves freely	Object provide data hiding gives important to data
3.	Overloading is not possible	Overloading is possible
4.	Inheritence ————	inheritance ————
5.	Code reusability does not present	Code reusability is present
6.	Ex : Pascal, C, FORTRAN etc.	Ex, C#, Java, C++ Python etc.

Objects are real word entity and has 2 things

- properties or State
- Behavior or function

Ex: Car is an object because it has properties like: color, Type, Brand, weight
Behavior like: Apply break, Drive, increase speed.

To creat a object class is required so class provide the template or Blueprint from which object can be created.

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1st Pillar of OOPs - Data Abstraction

- It hides the internal implementation and shows only essential functionality to the other
- It can be achieved through interface and abstract classes.

Ex: car - we only shown the Break pedal and if we press it car speed will reduce - But How] That is Abstracted to us.

Advantage of abstraction:
- Increase security and confidentiality.

2nd Pillar of OOPs - Data ~~Abstractio~~ Encapsulation

- Encapsulation bundles the data and the code working on that data in a single unit.
- Also known as Data-Hiding
- Steps to achieve encapsulation:
  - Declare variable of a class as private
  - Provide a public Getters and setters to modify and view the value of the variables

Advantages of Encapsulation
- loosely coupled code
- Better access control and security.

3rd pillar of OOPs – Inheritance

- capability of a class to inherit properties from their Parent class.
- It can inherit both functions and variables, so we do not have to write them again in child class.
- can be achieved using extends keyword or ~~through inherit~~ interface.

Types of inheritance

- Single Inheritance
- Multi level "
- Hierarchical "
- Hybrid "
- Multiple i. → through interface we can resolve the diamond problem.

Advantage of Inheritance

- code reusability

we can achieve polymorphism using inheritance.

4th pillar of OOPs – Polymorphism

- Poly means many and Morphism means form
- A same method, behaves differently in diff situation

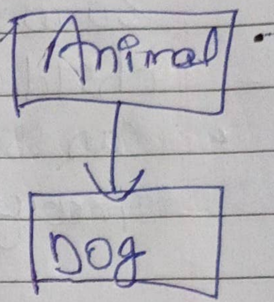
eg

- A person can be father, husband, employee etc
  water can be solid, liquid & gas etc.

Types of polymorphism

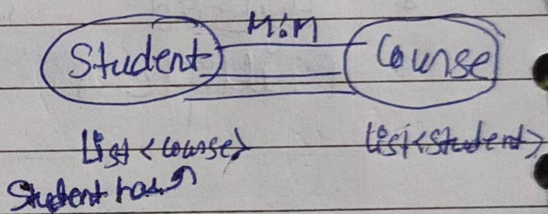
- Compile time / Static polymorphism / Method overloading
- Run time / Dynamic polymorphism / Method overriding

Object Relationships



- Is-a Relationship
  - Achieved through inheritance
  - Ex Dog is a Animal

- Has-a Relationship
  - whenever, an object is used in other class, it is called Has-a relationship
  - Relationship could be one to one, One to many, many to many



Ex School has Students
Bike has Engine
School has Classes

Association: relationship b/w 2 diff objects

weak rel^n - Aggregation: Both obj can survive individually, means ending of one obj. will not end other object

Strong rel^n - Composition Ending of one object will end other object